

BYUNG HYUN LEE

CONTACT INFORMATION

Affiliation: Department of Electrical and Computer Engineering, Seoul National University

 Email  Website  Google Scholar  LinkedIn

RESEARCH INTERESTS

My research focuses on *selective knowledge updates* for deep visual and foundation models toward agentic/physical AI — adding new knowledge or removing unwanted knowledge while preserving existing capabilities. I have studied this through continual learning, unlearning, and reinforcement learning, which are essential for agentic/physical AI, vertical-AI, and visual generative AI. I am currently extending these ideas to multimodal LLMs, omnimodal models, vision-language-action models, and domain-specialized models, aiming at adaptive and reliable knowledge updates that efficiently scale to real-world settings. Additionally, I am interested in model acceleration, image restoration, and histopathology analysis.

EDUCATION

- **Seoul National University (SNU)** *Mar 2021 - Feb 2027 (expected)*
Combined M.S./Ph.D Program in Electrical and Computer Engineering (ECE)
- **Ulsan National Institute of Science and Technology (UNIST)** *Mar 2015 - Aug 2018*
B.S. in Electrical Engineering, GPA: 4.04/4.30

RESEARCH ADVISOR

- **Se Young Chun:** Professor, Department of Electrical and Computer Engineering (ECE), SNU

PUBLICATIONS (Top AI Conference Papers)

- [Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs](#)
H. Seo*, **B. H. Lee***, M. Kim*, D. Mah, J. Lee, S. Y. Chun (*co-first authors)
Conference on Neural Information Processing Systems (NeurIPS, Spotlight), 2026
- [Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models](#)
H. Seo*, **B. H. Lee***, J. Cho, S. Lim, S. Y. Chun
Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- [Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis](#)
B. H. Lee, W. Jeong, W. Han, K. Lee, S. Y. Chun
International Conference on Computer Vision (ICCV), 2025
- [Localized Concept Erasure for Text-to-Image Diffusion Models Using Training-Free Gated Low-Rank Adaptation](#)
B. H. Lee*, S. Lim*, S. Y. Chun (*co-first authors)
Conference on Computer Vision and Pattern Recognition (CVPR), 2025
- [Concept pinpoint eraser for text-to-image diffusion models via residual attention gate](#)
B. H. Lee*, S. Lim*, S. Lee, D. U. Kang, S. Y. Chun (*co-first authors)
International Conference on Learning Representations (ICLR), 2025

- [Doubly perturbed task free continual learning](#)
B. H. Lee, M. Oh, S. Y. Chun
Proceedings of the AAAI Conference on Artificial Intelligence (AAAI, Oral), 2024
- [Online Continual Learning on Hierarchical Label Expansion](#)
B. H. Lee*, O. Jung*, J. Choi, S. Y. Chun (*co-first authors)
International Conference on Computer Vision (ICCV), 2023
- [All-in-one image restoration for unknown degradations using adaptive discriminative filters for specific degradations](#)
D. Park, B. H. Lee, S. Y. Chun
Conference on Computer Vision and Pattern Recognition (CVPR), 2023

PUBLICATIONS (Journal Papers)

- [Continual Test-Time Adaptation for Robust Remote Photoplethysmography Estimation](#)
H. Lee, H. Lee, B. H. Lee, S. Y. Chun
IEEE Access, 2025
- [Towards accelerating model parallelism in distributed deep learning systems](#)
H. Choi*, B. H. Lee*, S. Y. Chun, J. Lee (*co-first authors)
PLOS One, 2023

TECHNICAL REPORTS / MANUSCRIPTS UNDER REVIEW

- [Hyperbolic Vision-Language Model for Pathological Patch-Slide Representation](#)
B. H. Lee*, J. H. Jang*, Y. Kim, H. Kim, S. Y. Jeon, J. Cho, J. Lee, H. Seo, H. Bae, S. Y. Chun
(*co-first authors)
Under review, 2026
- [Multi-student On-Policy Self-Distillation with Alternating Reinforcement Learning for Robust Vision-Language-Action Models](#)
H. Kim*, B. H. Lee*, J. Cho, J. Park, Y. Seo, Y. Hwang, D. Shin, S. Y. Chun (*co-first authors)
Under review, 2026
- [S³-OPSD: Sparse, Stable and Speedy On-Policy Self-Distillation](#)
H. Seo*, B. H. Lee*, M. Oh, S. Y. Chun (*co-first authors)
Under review, 2026
- [RePath: Zero-Shot Whole-Slide Tumor Segmentation via Intra-Slide Relational Affinity](#)
D. U. Kang*, S. Lim*, J. Cho, B. H. Lee, K. Lee, S. Y. Chun (*co-first authors)
Under review, 2026
- [Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis](#)
H. Seo*, H. Bae*, B. H. Lee*, J. Cho, J. Lee, J. H. Jang, H. Kim, S. Y. Jeon, Y. Kim, S. Y. Chun
(*co-first authors)
Under review, 2026
- [HyperCLOVA X 8B Omni](#)
HyperCLOVA X Team in NAVER Cloud (including B. H. Lee)
2026
- [HyperCLOVA X 32B Think](#)
HyperCLOVA X Team in NAVER Cloud (including B. H. Lee)
2026
- [Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation](#)
H. Seo*, J. Bang*, H. Lee*, J. Lee, B. H. Lee, S. Y. Chun
arXiv, 2025

PUBLICATIONS (Short, Workshop, or Domestic Papers)

- [Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models](#)
K. R. Lee*, K. H. Lee*, S. Hong, **B. H. Lee**, S. Y. Chun
CVPR Workshop on Machine Unlearning for Computer Vision, 2026
- [Selective Concept Erasing for Safe Diffusion Models](#)
B. H. Lee, S. Lim, S. Y. Chun
Korea Signal Processing Conference (Best Poster Presentation Award), 2024
- [Expert classifier ensemble based post-processing correction for unbiased scene graph generation](#)
S. Lee, **B. H. Lee**, S. Y. Chun
Workshop on Image Processing and Image Understanding (IPIU), 2024
- [Efficient and accurate quantized image super-resolution on mobile npus, mobile ai & aim 2022 challenge: report](#)
A. Ignatov et al. (including **B. H. Lee**)
European Conference on Computer Vision Workshops (ECCVW), 2022
- [Efficient single-image depth estimation on mobile devices, mobile AI & AIM 2022 challenge: report](#)
A. Ignatov et al. (including **B. H. Lee**)
European Conference on Computer Vision Workshops (ECCVW), 2022
- [Uncertainty-based dual domain low-dose X-ray CT reconstruction](#)
S. Lee, D. U. Kang, **B. H. Lee**, S. Y. Chun
Korean Signal Processing Conference, 2022
- [Empirically Accelerating Scaled Gradient Projection Using Deep Neural Network for Inverse Problems in Image Processing](#)
B. H. Lee, S. Y. Chun
International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2021

PATENTS

- **All-in-one image quality improvement model providing method performing image quality restoration for multiple image quality degradation factors**
S. Y. Chun, D. Park, **B. H. Lee**
U.S. Patent, Registered, 2026 (Filed, 2023)
- **Method for providing all-in-one image quality improvement model that performs image quality restoration for multiple image quality inhibitors**
S. Y. Chun, D. Park, **B. H. Lee**
Korea Patent, Registered, 2025 (Filed, 2023)

AWARDS AND HONORS

- **Compute Transparency Champion Award** 2026
CVPR 2026 (for “Erasing Thousands of Concepts”)
- **Outstanding Reviewer (Top 5%)** 2026
Conference on Computer Vision and Pattern Recognition (CVPR)
- **Apple Scholars in AI/ML PhD Fellowship Program** 2025
Institutional Nominee
- **Qualcomm Innovations Fellowship Korea** 2025
Best paper finalist
- **2nd Place in Report Generation in Pathology using Pan-Asia Gigapixel WSIs** 2025
Challenge by International Conference on MICCAI (Served as the team leader, 26 teams participated)
- **Yulchon AI Star Scholarships (₩8,000,000)** 2025
Youlchon Foundation & SNU AI Institute

PROJECTS

- **Uncertainty-Aware Bidirectional Interaction between World Models and Vision-Language-Action Models** *Jul 2026 - Present*
AI Research Computing Support Program, Korea Electronics Technology Institute (KETI), Republic of Korea
- **Hyperbolic Pathology Foundation Models for Gigapixel Report Generation** *Jan 2026 - Jun 2026*
AI Computing Resource (GPU) Support Program, MSIT & NIPA, Republic of Korea
- **AI for Human-Machine Teaming** *Aug 2025 - Present*
United States Air Force & IITP, Republic of Korea
- **Industrial Defect Segmentation** *Dec 2024 - Mar 2025*
Samsung Device Solution (DS)
- **AI System for Gigapixel Histopathological WSI Analysis** *Aug 2023 - Dec 2024*
Ministry of Health and Welfare, Republic of Korea
- **Relations and Affordance Aware Robot Grasping** *May 2022 - May 2023*
Samsung Research, Republic of Korea

MEDIA COVERAGE

- **Popular Science Korea** *Jun 2026*
[Featured article on “Erasing Thousands of Concepts” \(CVPR 2026\), which erases 2,072 concepts from a text-to-image diffusion model](#)

EXPERIENCES

- **National AI Foundation Model Project in South Korea**, NAVER Cloud *Oct 2025 - Feb 2026*
Visiting Student Researcher
- **Biomedical Medical Image Processing Lab**, UNIST *May 2020 - Feb 2021*
Researcher (Advisor: Prof. Se Young Chun)
- **Biomedical Medical Image Processing Lab**, UNIST *July 2017 - Aug 2018*
Student Internship (Advisor: Prof. Se Young Chun)

PRESENTATIONS

- **Invited Poster Presentation on Selective Knowledge Updates for Deep Visual Models**
Samsung AI Forum (SAIF) 2025, Samsung Electronics, Yongin, Korea, Sep 2025
- **Continual Learning and Its Applications in Magnetic Resonance Imaging**
Advanced neuroimaging and AI workshop, SNU, 2024

ACADEMIC SERVICES

- **Reviewer**: NeurIPS, CVPR, ECCV, AAAI, IEEE Transactions on Image Processing (TIP), Pattern Recognition

EXTRACURRICULAR EXPERIENCES

- **Military Service, Republic of Korea Army** *Sep 2018 - Apr 2020*
Discharged as Sergeant
- **Pinocchio (Robot Club)**, UNIST *Mar 2015 - Aug 2018*
Education Director